THE MAGNETIC PROPERTIES AND USES OF IRON ALLOYS

By R. D. HAIGH, M.Eng., Graduate.*

[ABSTRACT *of paper read before the* MERSEY AND NORTH WALES (LIVERPOOL) STUDENTS' SECTION *3rd May*, 1941.]

INTRODUCTION

The properties and uses of iron alloys are reviewed largely from the point of view of the communications engineer. Nickel-iron alloys form the most important group of soft magnetic materials considered, and brief mention is made of some permanent-magnet materials.

HISTORY AND DEVELOPMENT OF IRON ALLOYS

Nickel-iron alloys were first prepared synthetically by Stodart and Faraday in 1820, but the commercial preparation was not demonstrated until 1885 by Marbeau. The magnetic properties were not appreciated until 1921 when the Western Electric Co. of America took out a patent for an alloy containing 78 % nickel which had a higher permeability and lower losses than the best magnetic materials then available, namely pure iron and silicon-iron alloys. The first alloy commercially used by the company contained 78·5 % nickel and 21·5 % iron and is an outstanding member of the group of alloys containing between 30 % and 95 % nickel and known as permalloys. Two years later Smith and Garnett took out patents for alloys having similar properties to the permalloys but containing copper and manganese in addition to nickel and iron. The most important alloy in this series is mumetal. Both mumetal and 78·5-permalloy are characterized by high permeabilities at low magnetizing forces, thus rendering them particularly suitable for communications work. Mumetal has a rather lower permeability than 78·5-permalloy but has a higher electrical resistivity, which is of advantage in reducing eddy-current losses.

The Bell Technical Laboratories in America pioneered research into the whole realm of combinations of nickel, iron and cobalt. The addition of cobalt aims at further reducing the hysteresis loss.

Desirable properties for communications work are high permeability, and low hysteresis and eddy-current losses, and in many cases constancy of permeability over the operating range. High electrical resistivity is necessary in order to reduce eddy-current losses, and the material should therefore be suitable for manufacture in the form of thin laminations. The Bell Telephone Laboratories found that the main factors affecting the final characteristics of the materials are the purity of the elements used, their preparation, and the heat treatment. The heat treatment is particularly important, as the alloys are very susceptible to machining stresses, and the final annealing should therefore be performed after all machining is completed.

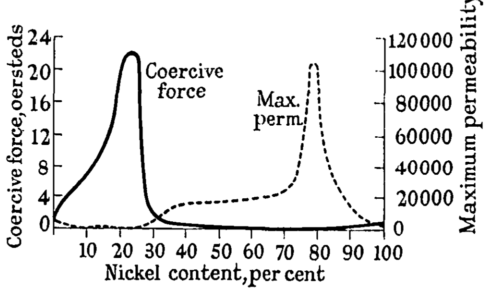

Fig. 1.—Variation of permeability and coercive force of nickel-iron alloys as nickel content increases.

Fig. 1 shows the variation in coercive force and permeability of nickel-iron alloys as the nickel content varies. The alloys containing more than 50 % nickel require rapid cooling in order to develop the magnetic qualities. Their low coercive forces indicate low hysteresis losses.

PROPERTIES OF IRON ALLOYS

The magnetic properties of the main types of alloys compared with those of pure magnetic iron are set out in Table 1.

Table 1

PROPERTIES OF SOFT MAGNETIC MATERIALS

Material	Initial permeability	Maximum permeability	Saturation flux	Resistivity
			lines/cm^2	microhm-cm.
Magnetic iron	250	5 500	21 500	10
45-permalloy	2 700	23 000	16 500	45
78·5 permalloy	9 000	105 000	10 700	16
Mumetal	7 000	80 000	8 500	25
4–79 molybdenum permalloy	22 000	72 000	8 500	55
3·8–78·5 chromium permalloy	10 000	40 000	8 000	65
45–25 perminvar	365	1 800	15 500	19
7·5–45–25 molybdenum perminvar	550	3 800	10 300	80
Permendur	800	5 000	24 500	7
2V-permendur	800	4 500	24 000	26

* Automatic Telephone and Electric Co., Ltd.

When other metals are added to permalloys their resistivity is generally increased. This is illustrated in the case of the chromium and molybdenum permalloys. It is also remarkable that the addition of 4 % of molybdenum (a non-magnetic metal) to 79-permalloy should increase the initial permeability from 9 000 to 22 000.

The perminvars are alloys containing manganese, cobalt, nickel and iron. They are characterized by constancy of permeability at low flux densities, a low hysteresis loss in this range and, for medium flux densities, a constriction in the middle of the hysteresis loop. If 45–25 perminvar is heat-treated by baking for 24 hours at 425° C. the constriction becomes so extreme that the coercive force vanishes and the two branches of the loop coincide when the magnetizing force is reduced to zero (see Fig. 2). The

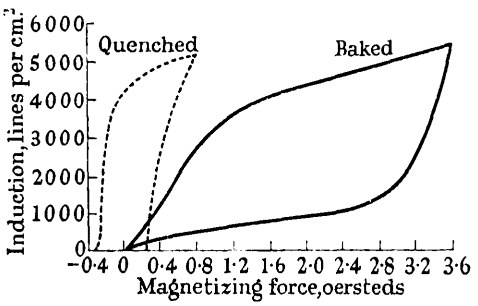

Fig. 2.—Hysteresis loop for 45–25 perminvar.

dotted line shows the same alloy in a quenched condition when the constriction has almost disappeared. The constriction also disappears if the loop is taken at high flux densities (10 000 lines/cm² or more).

The explanation of the constriction is not certain, but it appears probable that it may be caused by segregation of the alloy—a phenomenon more likely to occur in baking than in quenching. This theory is also supported by the fact that a bimetallic parallel magnetic circuit gives a similar constriction, e.g. a core of piano wire surrounded by a permalloy tube.

The use of perminvar at higher frequencies is rather limited by its low resistivity, but, as will be seen from Table 1, the addition of 7·5 % molybdenum greatly increases the resistivity.

An alloy of 50 % cobalt and 50 % iron is known as "permendur" and is noteworthy for its high saturation flux and low resistivity. It is brittle and therefore difficult to roll into thin sheets, but the addition of 2 % vanadium reduces the brittleness and also increases the resistivity fourfold.

PERMANENT-MAGNET MATERIALS

Table 2 gives the properties of some typical materials. Alnico and mishima are outstanding among the steels. Alnico contains 10 % aluminium, 6 % copper, 17 % nickel, 12·5 % cobalt and 54·5 % iron; and mishima 13 % aluminium, 29 % nickel and 58 % iron. Permanent-magnet materials have reached a very successful stage from the magnetic point of view, but the greatest handicap of the better materials is their extreme hardness, which complicates fabricating processes.

Table 2

PROPERTIES OF MAGNETIC STEELS

Material	Permeability	Residual magnetism	Coercive force
		lines/cm²	oersteds
Manganese steel ..	75	10 000	50
3 % chromium steel	31	9 700	65
5 % tungsten steel	32	10 000	60
36 % cobalt steel ..	7	9 500	220
Alnico steel ..	4	7 200	540
Mishima steel ..	4	6 000	550

APPLICATIONS OF MAGNETIC MATERIALS

Relays

The normal material for telephone-relay cores is Swedish iron, but where great sensitivity or special release conditions are required the additional expense of permalloy is justified. A further use of permalloy is where the relay coil shunts the line as in a transmission bridge and high inductance is necessary in order to reduce voice-frequency losses. In this case one or more permalloy sleeves are put over the soft-iron core.

Transformers

The low losses and high permeabilities of the permalloys make them especially suitable as core materials for transformers in telephone circuits used to transmit music. In addition to improved transmission qualities the use of permalloy instead of iron substantially reduces the size of the transformer.

Loading Coils

Up to 1926 the cores of loading coils for telephone lines were made of finely divided iron, the particles being insulated from one another by shellac varnish and other materials. Subsequently experiments were made with permalloy, and its high electrical resistivity, low hysteresis loss and coercive force and high permeability soon showed it to be particularly suitable. Fig. 3 shows the resulting

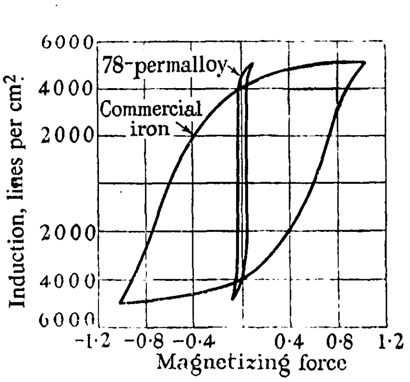

Fig. 3.—Hysteresis loops of iron and 78-permalloy.

decrease in hysteresis loss. The permalloy was first used in the form of cropped wire suitably insulated and compressed, but present-day cores employ permalloy dust.

Considerable reduction in size was effected by the use of permalloy, and the Americans have recently introduced a molybdenum permalloy which practically halves the size again for a given inductance.

Magnetostrictional Effects and Uses

Of the high-permeability materials 45-permalloy gives the largest magnetostrictional effect. If a rod of the material is placed in an alternating field it will resonate when the frequency is in tune with the natural frequency of longitudinal vibration of the rod. It thus forms a useful frequency standard.

The converse effect, in which an e.m.f. is generated in a coil when the enclosed core is vibrated, becomes objectionable as a source of background noise in high-gain amplifiers. Thus for such purposes an alloy with the minimum magnetostrictional effect, such as 81-permalloy or 4–79 molybdenum permalloy, must be chosen.

Crystal Filters

The initial permeability of the usual magnetic materials increases to a maximum with increase of temperature, and then drops very rapidly as the temperature reaches the Curie point. In crystal filters where only minute variations in the resonant frequency are allowable this initial increase of inductance is objectionable. It would, in fact, be desirable that the inductance should fall slightly with increase in temperature to counterbalance the slight increase in capacitance of mica condensers with increase in temperature.

This is done by the addition to the usual 2–81 molybdenum permalloy of a small amount of permalloy containing $12 \cdot 5\,\%$ molybdenum. The latter material has a Curie point only just above room temperature, and thus a large negative temperature coefficient of inductance. The net result of the mixture is therefore to give a material with a small negative coefficient.

Miscellaneous Uses

The many other uses of iron alloys include moving-coil microphones, magnetic-tape recording-machines, magnetos and choke coils, all of which can be improved in efficiency by the judicious use of modern materials.

CONCLUSIONS

The engineer has now such a wide selection of magnetic materials from which to choose that a suitable alloy can be found for almost any conceivable purpose. The cost of many of these alloys precludes their use for general purposes, so that iron and silicon steel are in no danger of being ousted. The additional cost of special alloys may be outweighed by economies of space, weight, etc., and many such cases remain to be explored.

Further improvements in soft magnetic materials are limited partly by the difficulties of commercial application to laboratory techniques and partly by the ultimate limitations of the materials themselves.

Improvements in permanent-magnet materials are limited mainly by difficulties of fabricating the extremely hard materials which are most suitable magnetically.